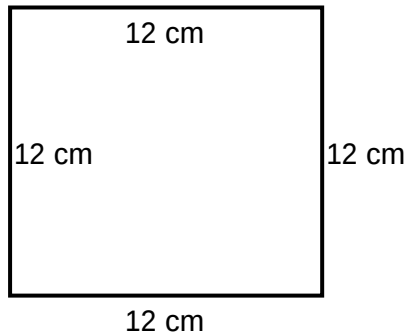


- PERIMETER

The perimeter of a shape is the total length around the outside of the shape. This distance is found by adding together the lengths of all the sides.

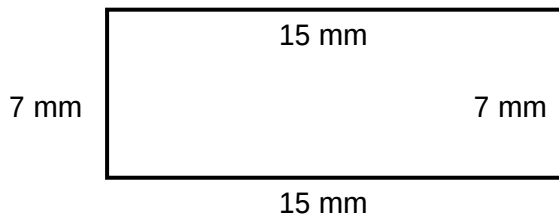
Example 1



A square where each side measures 12 cm.

$$\text{Perimeter} = 12 + 12 + 12 + 12 = 4 \times 12 = 48 \text{ cm}$$

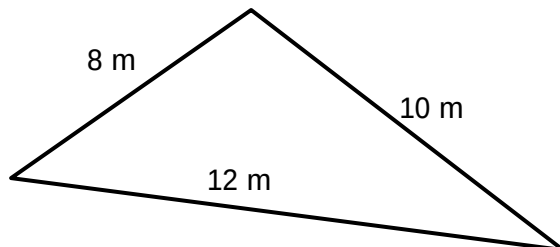
Example 2



A rectangle with a length of 15 mm and a width of 7 mm.

$$\text{Perimeter} = 15 + 7 + 15 + 7 = 2 \times 15 + 2 \times 7 = 44 \text{ mm}$$

Example 3



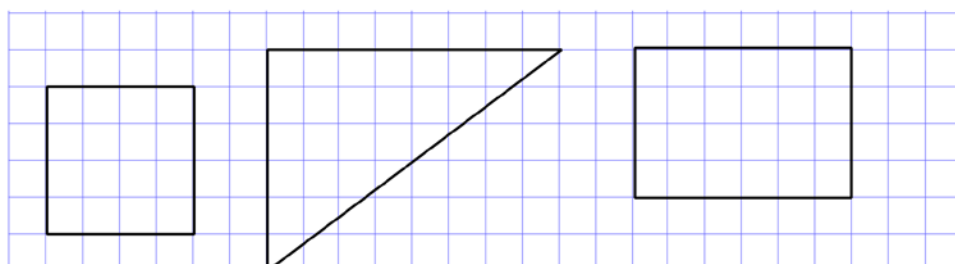
A triangle with three sides measuring 8 m, 10 m and 12 m.

$$\text{Perimeter} = 8 + 10 + 12 = 30 \text{ m}$$

Exercise 3.2

Perimeter

1. Use a ruler to measure the perimeter of these shapes to the nearest centimetre.



Exercise 3.2 continued

2. Using a ruler, find the perimeters of the following rectangles, to the nearest centimetre.

a)



b)



c)



d)



e)

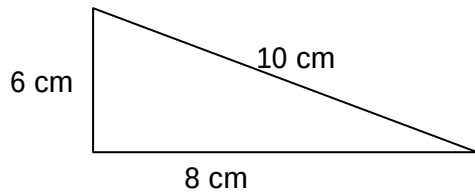


f)

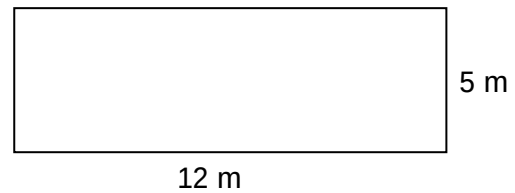


3. Find the perimeter of the following shapes, using the measurements given:

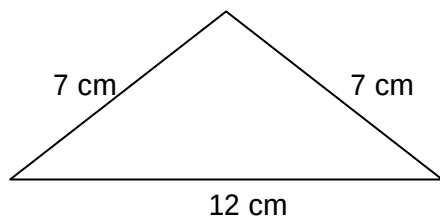
a)



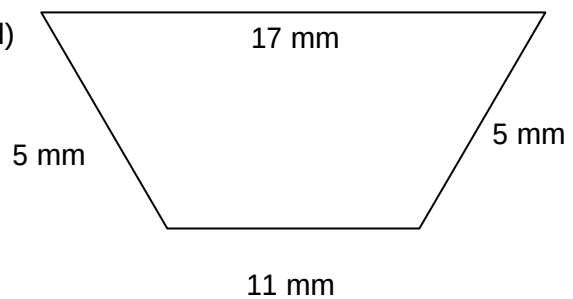
b)



c)



d)

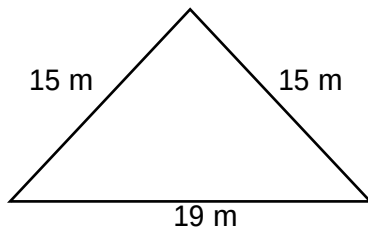


4. Use the dimensions given in the table to calculate the perimeters of the following squares and rectangles.

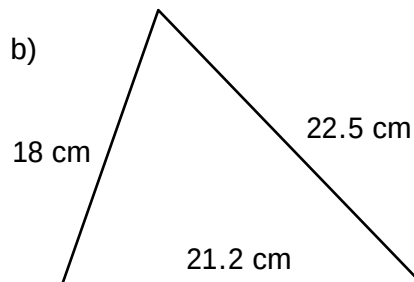
LENGTH	WIDTH	PERIMETER
4 m	3 m	
10 m	10 m	
12 cm	2 cm	
50 mm	20 mm	
1.5 m	3.5 m	
16 cm	8.5 cm	
7 km	7 km	

5. Calculate the perimeter of each triangle.

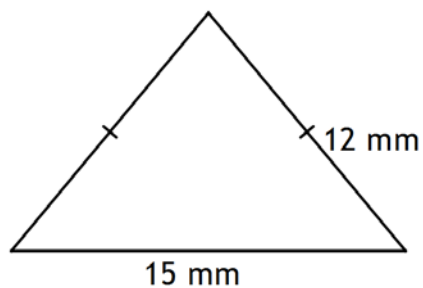
a)



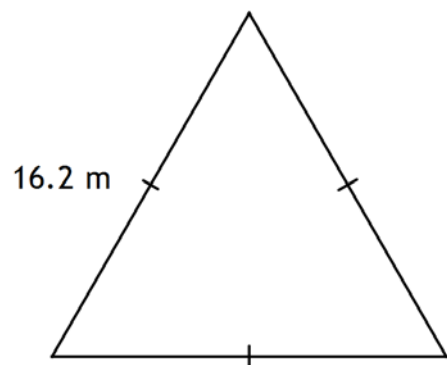
b)



c)



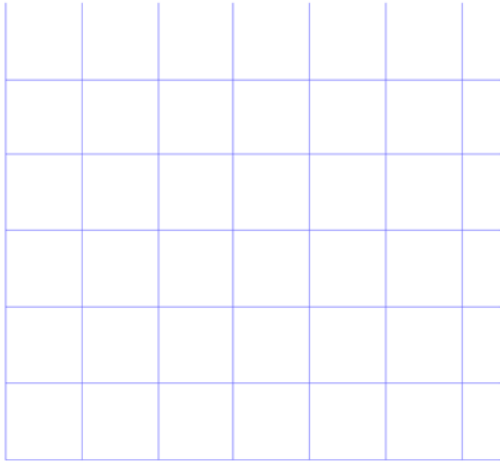
d)



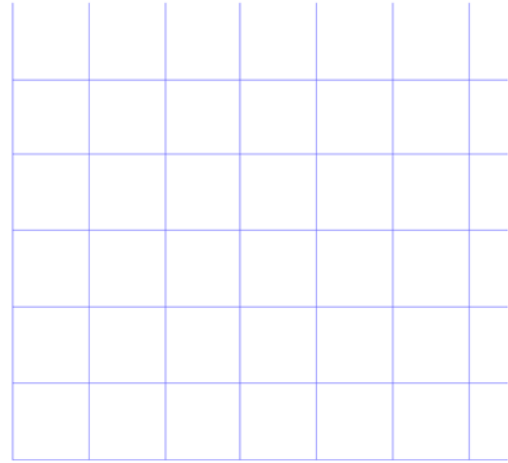
Exercise 3.2 continued

6. Draw squares with the given perimeters:

a) Perimeter = 8 cm



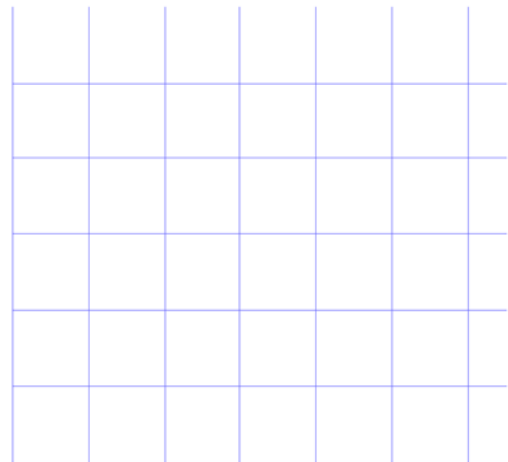
b) Perimeter = 12 cm



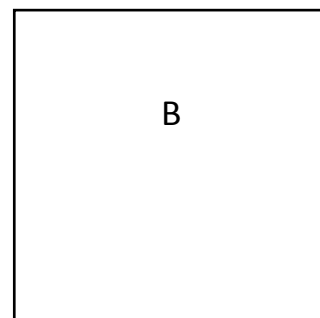
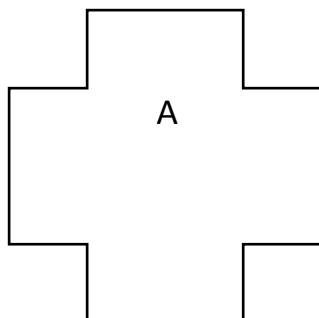
c) Perimeter = 4 cm



d) Perimeter = 16 cm



7. Which of these figures has the greatest perimeter?



8. Becky has a rectangular garden 22.4 m long and 14 m wide.

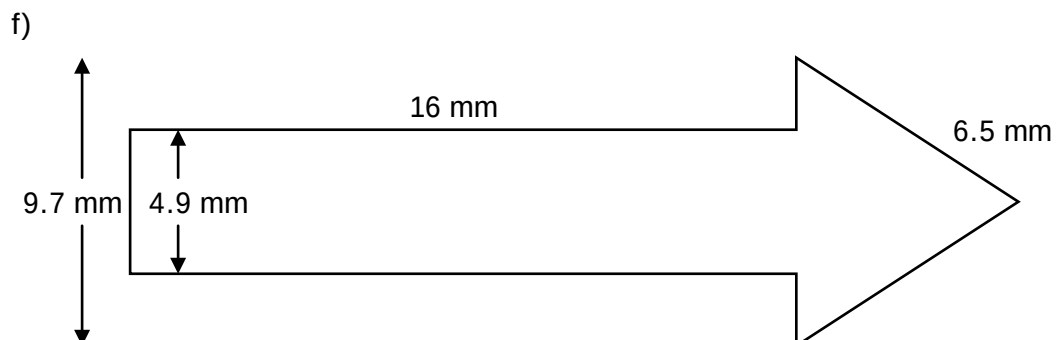
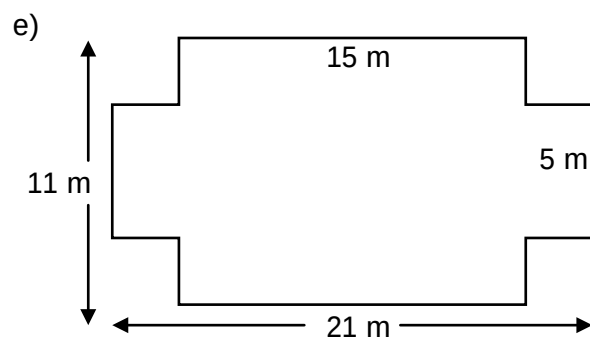
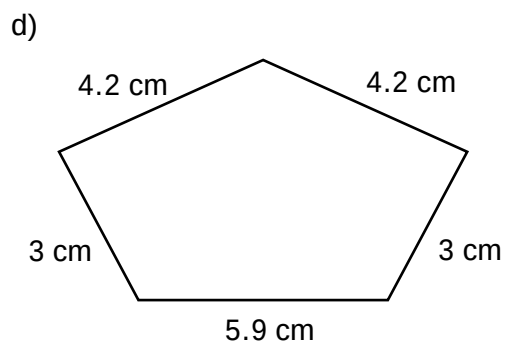
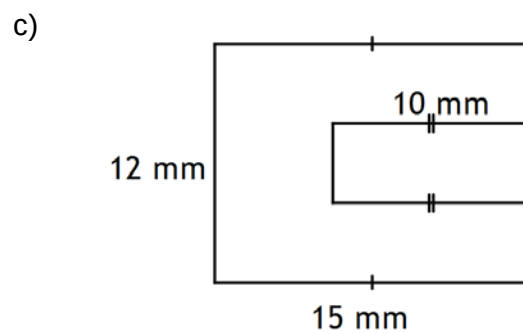
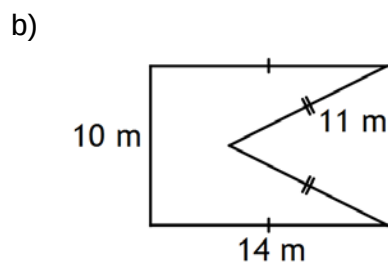
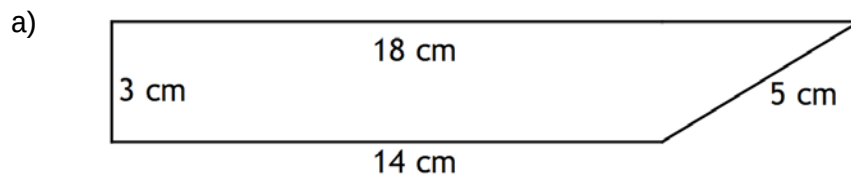
a) Each morning she walks around her garden twice. How far does she walk?

She decides to plant a 3 m wide strip of lawn all the way around her garden and put a fence around the lawn.

b) How long will the fence be?



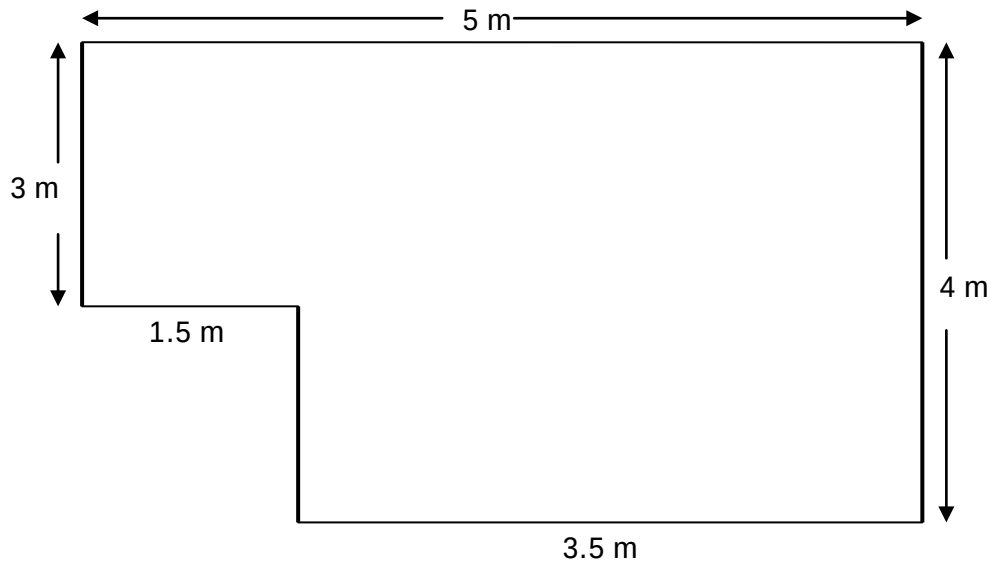
9. Find the perimeter of the following shapes.



Exercise 3.2 continued

10. An L-shaped room is to have new skirting boards.

The plan of the room is shown but it is not drawn to scale.



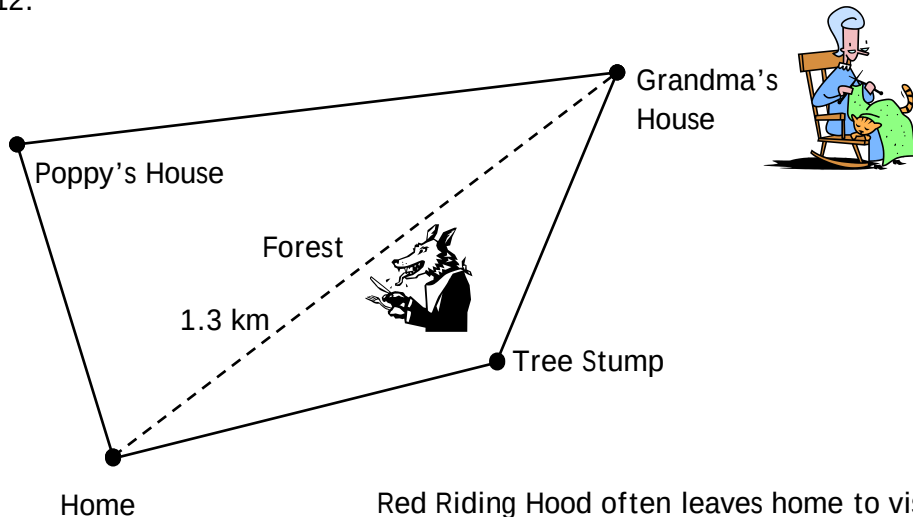
- a) Find the perimeter of this room.
- b) What would the skirting boards cost if they are \$42 per metre?
11. Bella has three rectangular pictures to frame. Two of them are the same size, 18 cm by 32 cm, while the other is larger, 50 cm by 80 cm.
- a) What is the perimeter of the smaller pictures? the larger picture?

The cost of the framing material is \$14 per metre with a single \$35 labour fee added.

- b) How much will it cost to frame the three pictures?



12.



Red Riding Hood often leaves home to visit Grandma or her friend Poppy. The distance to Grandma's house directly through the forest is 1.3 km, but if she walks around the forest, via the tree stump, the distance to Grandma's house is 2.4 km. Poppy's house is 2.2 km from Grandma's house.

On Wednesday she leaves home and walks to Grandma's house via the tree stump, then on to Poppy's house and back home. She travels a total distance of 5.5 km.

a) How far is Poppy's house from Red Riding Hood's house?

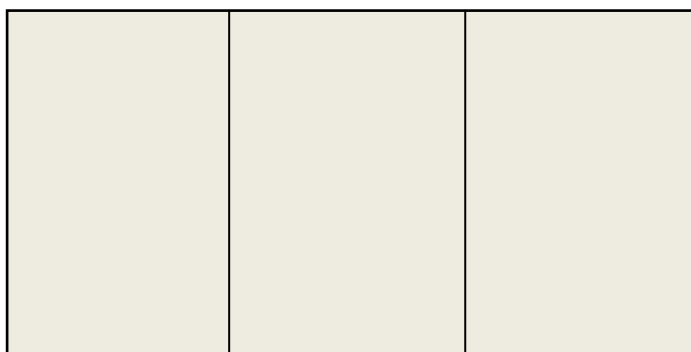
On Friday she visits Poppy first then Grandma and returns home through the forest.

b) How much further did she walk on Wednesday than on Friday?

13. A netball court is 30.5 m long and 15.25 m wide.

a) Find the perimeter of the court.

The court is divided into thirds with two more lines, known as transverse lines, painted across the court as shown:



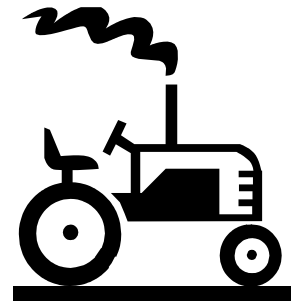
b) How much would it cost to mark out the boundary of the court plus the two transverse lines, at \$1.58 per metre?

Exercise 3.2 continued

14. The perimeter, or circumference, of a circular tractor wheel is 1.45 m.

The tractor travels the length of a 120 m long paddock 14 times.

- How far has the tractor travelled?
- How many times has the wheel rotated?



15. Australian Rules Football grounds vary in length. Games played at the MCG are played on a field that is 160 m long, while games played at Paterson Stadium (formerly Subiaco Oval) are played on a field that is 175.6 m long.

Bobby Runalot runs, on average, 40 lengths of the field when he gets a game.



- How much further would he run during a game at Paterson Stadium than at the MCG?

At training, the coach often makes Bobby run around the perimeter of the ground 11 times. At the MCG the perimeter of the ground is approximately 585 m, while at Paterson Stadium the perimeter is approximately 602 m.

- How much further would he run during training at Paterson Stadium than at the MCG, in metres? in kilometres?

16. To run a track and field event requires many ground markings. A competition including a 100 m sprint needs 8 lanes, each being 1.22 m wide. A 3 m start box is also required and a 17 m run-out distance.



On the diagram, which is not drawn to scale, write in all the distances given and work out the total length of all the marking lines shown. Give your answer in metres, and in kilometres.